PROJECT PROPOSAL

Date of proposal: 30/09/2026

Project Title: Multimodal Emotion Recognition Using Speech and Facial Expressions

Group ID (As Enrolled in Canvas Class Groups): Group 13

Group Members (name, Student ID):

Agarwal Hritik Kumar A0353874R
Jiang Yushan AX0353592X
Lu Yujing A0351286B
Ma Bofei A0314662H
Zhang Yurun A0330891A

Sponsor/Client: *(Company Name, Address and Contact Name, Email, if any)*
N/A

Background/Aims/Objectives:

Human communication conveys not only linguistic content but also emotional information. The same spoken sentence can express different emotions through variations in tone, pitch, intensity, and speaking style, while facial expressions can provide additional emotional cues. Therefore, automatic emotion recognition from speech and related visual information is an important task in affective computing and human–computer interaction.

In this project, the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is used to investigate emotion recognition. The project focuses specifically on the speech subset of RAVDESS and excludes the song recordings. The selected speech data contain recordings from 24 professional actors and cover eight emotional categories: neutral, calm, happy, sad, angry, fearful, disgust, and surprise. Both audio and video information available in the speech recordings are considered.

The main objective of this project is to investigate whether combining speech and facial expression information improves eight-class emotion recognition on the RAVDESS speech subset under a speaker-independent evaluation setting. Specifically, the project aims to:

- Develop and evaluate audio-only and visual-only emotion recognition baselines using consistent actor-independent training, validation, and test partitions.
- Compare alternative deep-learning encoders for audio and visual emotion recognition under a consistent experimental protocol.
- Compare decision-level fusion, feature concatenation, and gated fusion, with cross-attention explored as an optional extension, to determine whether multimodal approaches outperform the strongest single-modality baseline.
- Assess performance primarily using macro-averaged F1-score, supported by

accuracy and confusion matrices, to identify overall performance differences and commonly confused emotion categories.

Project Descriptions:

1 Data Preparation and Partitioning

1.1 Dataset Overview

Attribute	Description
Dataset	RAVDESS
Selected subset	Speech
Number of actors	24
Gender	12 male, 12 female
Unique speech trials	1,440
Total selected files	4,320
Audio-only files	1,440
Audio-Visual (AV) files	1,440
Video-only (VO) files	1,440
Emotion classes	8
Emotional intensity	Normal / Strong*
Statements	2*
Repetitions	2
Audio format	16-bit, 48 kHz WAV
Video format	720p H.264 / AAC, 48 kHz
Language	North American English

Fig 1.1 Dataset Overview

*Neutral emotion does not have a strong-intensity condition.

*Statement (01 = "Kids are talking by the door", 02 = "Dogs are sitting by the door").

1.2 Emotional Distribution

Emotion	Number of trials
Neutral	96
Calm	192
Happy	192
Sad	192
Angry	192
Fearful	192
Disgust	192
Surprise	192
Total	**1,440**

Fig 1.2 Emotional Distribution

1.3 Data Selection and Organization

The original RAVDESS dataset contains both speech and song recordings. For this project, only speech recordings are retained, while all song recordings are excluded. For the selected speech subset, the audio-only files and the corresponding audio-visual and video-only files are organized according to actor and trial identifiers.

The RAVDESS filename convention provides information about modality, vocal channel, emotion, intensity, statement, repetition, and actor identity. These identifiers are used to organize the files and obtain the corresponding labels.

Here is a filename example: *02-01-06-01-02-01-12.mp4*

1. Video-only (02)
2. Speech (01)
3. Fearful (06)
4. Normal intensity (01)
5. Statement "dogs" (02)
6. 1st Repetition (01)
7. 12th Actor (12)
8. Female, as the actor ID number is even.

1.4 Basic Pre-processing

Before model training and evaluation, the selected data are checked for file integrity and organized into a consistent structure. Audio and video data are processed according to their respective formats, and basic normalization and formatting steps are applied where necessary. The same preprocessing principles are maintained across the training, validation, and test sets to ensure a consistent comparison.

1.5 Data Partitioning

To reduce the potential influence of speaker-specific characteristics, the data are partitioned at the actor level rather than randomly splitting individual files. This ensures that recordings from the same actor are kept within the same subset and that no actor appears in both the training and test sets. As a result, the models are evaluated on speakers they have not seen during training, providing a more reliable assessment of their ability to generalize to unseen speakers and recognize emotion-related patterns rather than speaker-specific characteristics.

Set	Actors	Trials	Approx. proportion
Training	16 actors	960	66.7%
Validation	4 actors	240	16.7%
Test	4 actors	240	16.7%
Total	**24 actors**	**1,440**	**100%**

Fig 1.3 Data Partitioning

2 Audio Emotion Recognition

This section aims to develop an Audio Emotion Recognition module using speech signals from the RAVDESS dataset. The module will identify speakers' emotions from their vocal

characteristics and provide audio feature representations and classification results for subsequent Multimodal Fusion. The overall workflow is shown in Figure 2.1.

First, we will apply a consistent preprocessing procedure to the original .wav audio files in RAVDESS. This will include resampling the audio to a common sampling rate (e.g., 16 kHz), amplitude normalization, trimming silence from the beginning and end of each recording, and basic denoising where necessary. These steps are intended to reduce recording differences and noise unrelated to emotion. We will then apply data augmentation only to the training set, using additive noise (jittering), time-warping, and amplitude scaling. These techniques will increase the diversity of the training samples while preserving their original emotion labels, with the aim of improving the model's generalization ability.

After preprocessing, we will use two complementary approaches in the Feature Extraction (Audio Representation) stage. The first approach will extract hand-crafted features by automatically computing features with clear physical meanings from each speech recording using established signal-processing methods. In the time domain, we will extract Zero-Crossing Rate (ZCR), Root Mean Square (RMS), and energy to describe waveform variations and sound intensity. In the frequency domain, we will use the Fast Fourier Transform (FFT) to transform the signal from the time domain to the frequency domain, then extract spectral centroid, spectral bandwidth, spectral energy, and spectral rolloff to describe how speech energy is distributed across frequencies. In addition, we will use the Discrete Wavelet Transform (DWT) to decompose the speech signal at multiple scales and calculate statistics such as mean, variance, and energy from different wavelet subbands. DWT can describe the temporal and spectral characteristics of speech at different scales, making it suitable for extracting features relevant to speech emotion recognition (Daqrouq et al., 2024). These time-domain, frequency-domain, and time-frequency features will be combined into a fixed-length hand-crafted feature vector.

The second approach, Step 4(b) *Learned Feature* in Figure 2.1, will use learned features, allowing a neural network to automatically learn more abstract representations of emotion from speech data. We will first extract Mel-Frequency Cepstral Coefficients (MFCCs) while retaining their variation over time. Each speech recording will be represented as a T × D MFCC sequence, where T is the number of time frames and D is the number of MFCC features per frame. The same MFCC sequence input will be used to compare three alternative Audio Encoders: a lightweight 1D-CNN, an LSTM/BiLSTM, and a Transformer Encoder. These models will be evaluated separately, not connected in series. Previous speech emotion recognition research has also combined MFCCs with a 1D-CNN to learn additional emotion-related features from MFCC inputs (Mary Little Flower et al., 2024). In our model, 1D convolution will operate along the time dimension to learn local patterns of change between neighboring speech frames, thereby extracting high-level features associated with different emotions.

The LSTM encoder will process MFCC frames sequentially, updating its hidden states and memory to capture longer temporal dependencies. A bidirectional LSTM (BiLSTM) will process the sequence in both directions, combining past and future context for each frame. The Transformer Encoder will first project each MFCC frame to a common embedding dimension and add positional encoding to preserve temporal order. Multi-head self-attention will then relate frames across the sequence, while feed-forward layers will transform these contextual features to model global relationships across time.

We will retain each Audio Encoder's sequence-level representation and use global pooling across time to obtain a fixed-length learned pooled feature representing the recording. The sequence will remain available before pooling; its temporal length may be reduced by the

CNN, whereas LSTM/BiLSTM and Transformer outputs will retain one representation per input frame.

In the Feature Fusion stage, we will concatenate the hand-crafted feature vector with the selected encoder's learned pooled feature obtained through global pooling to form the final pooled audio representation. This will combine manually designed signal-processing features with high-level features learned automatically by the neural network. The fused audio representation will then be passed to a multilayer perceptron (MLP) classifier. Its fully connected layers will learn the relationships between the audio features and different emotions, and a Softmax output layer will produce a probability distribution over eight emotion classes to complete Audio Emotion Recognition.

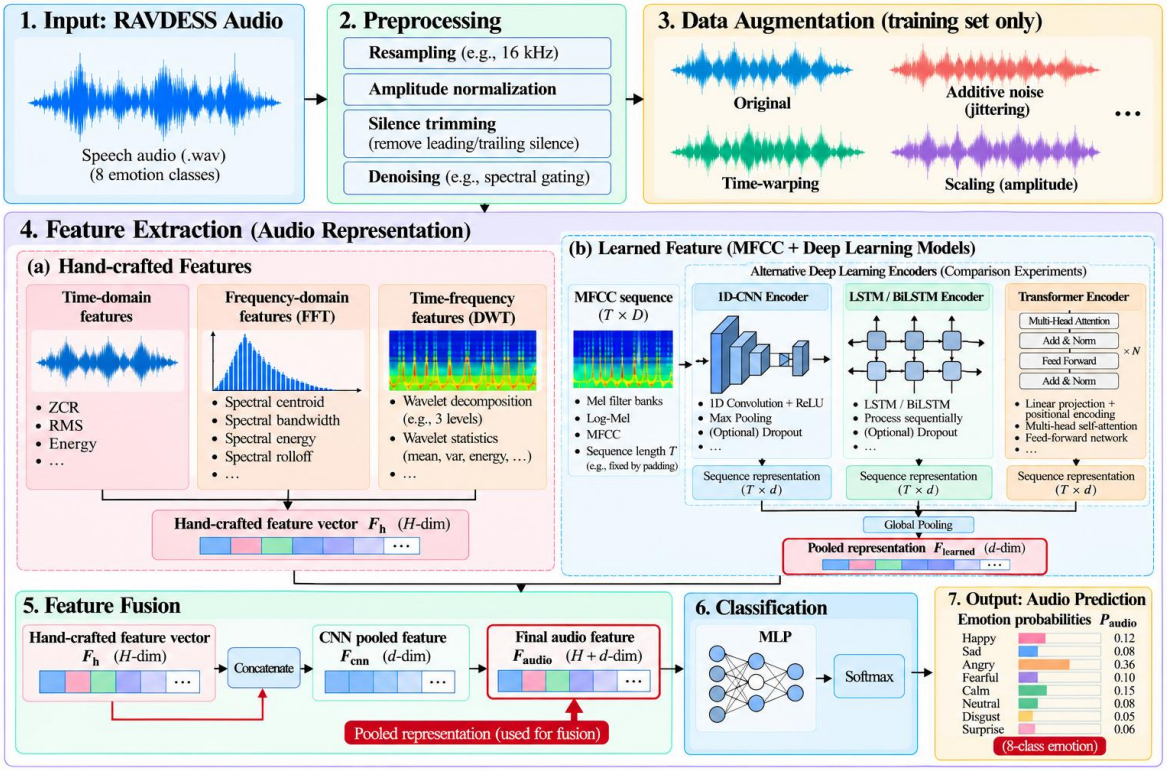


Fig 2.1 Audio Emotion Recognition Architecture

The experiments will have two levels. First, we will compare 1D-CNN, LSTM/BiLSTM, and Transformer encoders using the same data splits, MFCC inputs, preprocessing, training-only augmentation, and evaluation conditions. Each encoder's pooled feature will feed an MLP and eight-class Softmax classifier under a common training and evaluation protocol. We will select the best encoder based on validation F1-score, keeping the test set separate from model selection.

Second, to evaluate the contribution of different audio representations, we plan to conduct comparative experiments on the eight-class classification task using hand-crafted features only, learned features only from the selected encoder, and combined hand-crafted + learned features. Using the same data splits and evaluation protocol for all three feature configurations, we will compare their performance using Accuracy, Precision, Recall, and F1-score. These experiments will help us assess whether traditional signal-processing features and deep learned features provide complementary information, and whether combining them can improve Audio Emotion Recognition performance. Finally, the Audio module will provide the selected encoder's sequence-level representation, the final pooled representation formed by concatenating the hand-crafted vector with the selected learned

pooled feature, and 8-class emotion probabilities for subsequent Multimodal Fusion.

3 Visual Emotion Recognition

Facial expressions in the full audio-video speech recordings of RAVDESS will be analysed to distinguish eight emotions: neutral, calm, happy, sad, angry, fearful, disgust, and surprise (Livingstone & Russo, 2018). The visual pipeline covers frame sampling, face preprocessing, training-only augmentation, hand-crafted and learned feature extraction, temporal aggregation, two-level model comparison, and classification, as illustrated in Figure 3.1. The selected visual configuration will produce a single eight-class probability vector P_{visual} while retaining the representations required for multimodal fusion.

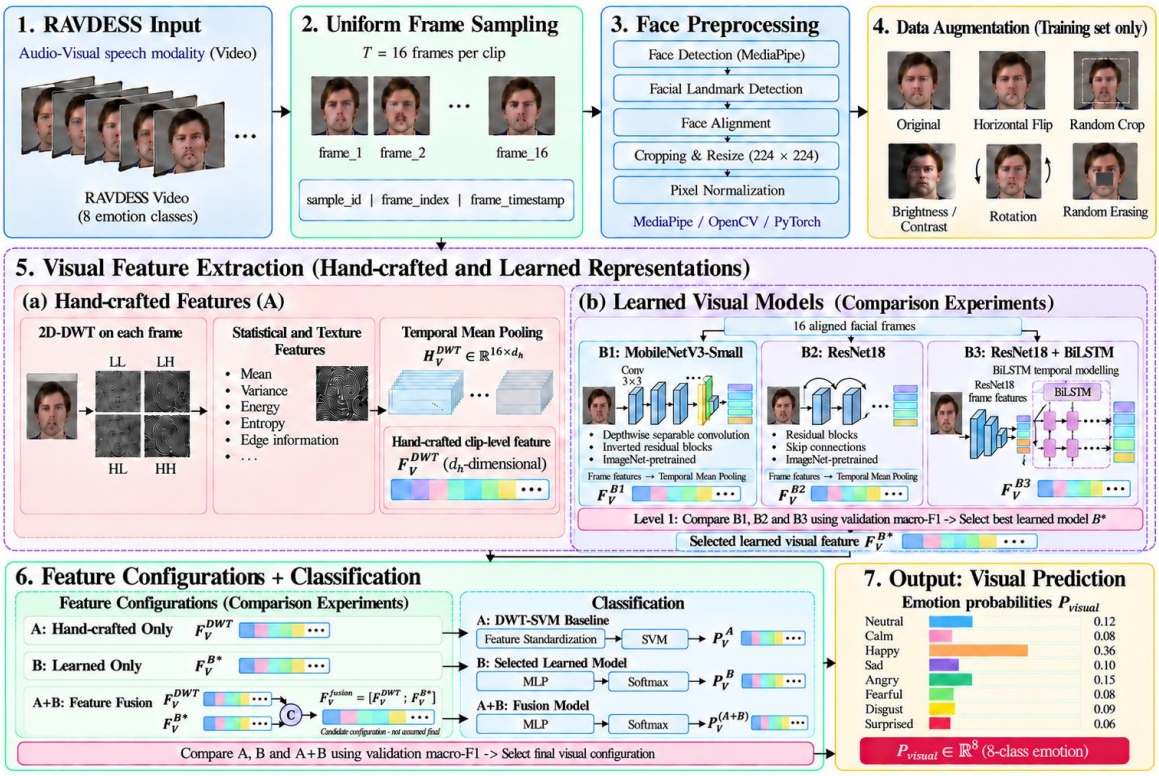


Figure 3.1. Visual Emotion Recognition Architecture

A fixed-length sequence will be created by uniformly sampling 16 frames across the full duration of each video clip. The number of sampled frames may be adjusted according to computational cost and validation performance. Faces and facial landmarks will then be detected to support alignment and cropping. Each resulting facial image will be resized to 224 × 224 pixels and normalized before feature extraction, reducing variations in background, facial position, scale, and orientation. MediaPipe will initially be used for face and landmark detection, OpenCV for alignment and image processing, and PyTorch transformations for tensor conversion and normalization (Lugaresi et al., 2019). Sample identifiers and frame timestamps will be preserved to maintain correspondence with the associated audio signals.

Training samples will be augmented through horizontal flipping, random cropping, moderate brightness and contrast adjustment, and optional small-angle rotation (Shorten & Khoshgoftaar, 2019). Geometric transformations will remain consistent across frames from the same clip so that augmentation does not introduce artificial temporal jitter. No random augmentation will be applied to the validation or test sets.

Two visual representation approaches will be examined. As a conventional signal-processing baseline, a two-dimensional Discrete Wavelet Transform will be applied to each

aligned facial image. The transformation will decompose the image into four subbands that represent the overall facial structure and the horizontal, vertical, and diagonal details at different spatial frequencies (Mallat, 1989). Statistical and texture descriptors, including mean, variance, energy, entropy, and edge information, will be extracted from these subbands and concatenated to form a hand-crafted feature vector for each frame. Mean pooling will then aggregate the frame-level vectors into a fixed-length representation of the video clip. After feature standardization, the resulting representation will be supplied to a Support Vector Machine for eight-class emotion classification, forming the DWT-SVM baseline.

For the learned representation approach, three learned visual configurations will be compared using the same 16 aligned facial frames, actor-independent data splits, preprocessing, training-only augmentation, and MLP-Softmax evaluation protocol. B1 will use an ImageNet-pretrained MobileNetV3-Small followed by temporal mean pooling. B2 will use an ImageNet-pretrained ResNet18 followed by temporal mean pooling. B3 will use ResNet18 to extract frame-level features and a BiLSTM to model their temporal development. The best learned configuration B* will be selected using validation macro-F1, with the test set kept separate from model selection.

For each learned configuration, the corresponding visual encoder will extract a feature vector from every aligned facial frame, producing a temporally ordered frame-level feature sequence $H_V^{B_i}$. For B1 and B2, temporal mean pooling will aggregate the frame-level sequence into a fixed-length clip-level feature $F_V^{B_i}$. For B3, the ResNet18 frame-level feature sequence will instead be processed by a BiLSTM to model changes in facial expressions over time and produce the clip-level representation $F_V^{B_3}$. Each learned representation will be passed to the same MLP classification head followed by a Softmax layer, producing an eight-class probability vector. Transfer learning will be adopted by initially freezing the earlier convolutional layers and training the classification head, followed by selective fine-tuning if this improves validation performance.

At Level 2, three complete visual configurations will be compared. Configuration A will use the temporally pooled DWT hand-crafted feature F_V^{DWT}, followed by feature standardization and an SVM classifier, forming the DWT-SVM baseline. Configuration B will use only the selected learned feature F_V^{B*}, followed by an MLP and Softmax classifier. Configuration A+B will concatenate the hand-crafted and selected learned features to form F_V^{fusion}, which will be passed to its own MLP and Softmax classifier. Feature fusion will therefore be treated as a candidate configuration rather than being assumed to be the final model. A, B, and A+B will be compared using the same data partitions and evaluation metrics, and the final visual configuration will be selected according to validation macro-F1. The selected configuration will produce a single eight-class visual probability vector $P_{visual} \in R^8$. The selected learned sequence, the clip-level representations, and P_{visual} will be retained for subsequent multimodal fusion with the corresponding audio outputs.

4 Multimodal Fusion

This component aims to integrate audio and visual representations for emotion

classification on RAVDESS. Previous research has explored decision-level fusion and cross-modal attention for combining speech and facial information (John & Kawanishi, 2022; Luna-Jiménez et al., 2021). As illustrated in Figure 4.1, we propose four alternative fusion pipelines using a common pair of audio and video encoders.

- Decision-level fusion. The class probabilities produced by the audio and video classifiers will be combined through weighted averaging. Equal weighting will serve as the initial configuration, followed by an adjustable weighting scheme to control the contribution of each modality.
- Feature concatenation. Audio and visual representations will be pooled and normalized separately, then concatenated and passed to a multilayer perceptron (MLP) for emotion classification. This provides a straightforward approach to learning a joint representation from the two modalities.
- Gated fusion. Inspired by gated multimodal units (Arevalo et al., 2017), a lightweight gating network will learn input-dependent fusion weights. Audio and visual features will first be projected into a common dimensional space and then combined using the learned gate, allowing their contributions to vary across samples.
- Cross-attention fusion. Subject to computational resources, a lightweight cross-attention module will be explored using audio and visual feature sequences. Audio features will act as queries, while visual features provide keys and values, enabling the model to learn cross-modal relationships. The resulting representation will be combined with the audio features through a residual connection, pooled, and passed to a classification head. This design draws on transformer-based audio–visual interaction (John & Kawanishi, 2022).

The four strategies will be implemented as alternative fusion pipelines, as shown in Figure 4.1. The audio and video encoders will initially remain frozen, allowing development to focus on the fusion modules. Both sequence-level and pooled representations will be retained to support the different fusion mechanisms. Each pipeline will independently produce a probability distribution over the eight emotion classes.

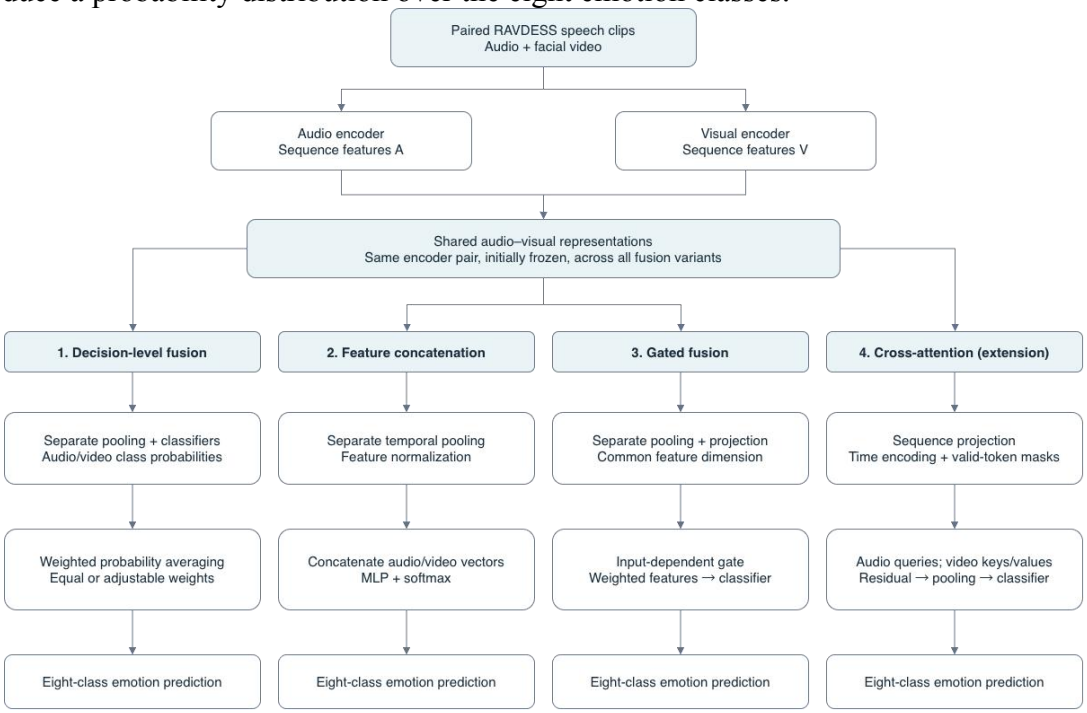


Figure 4.1 Experimental Design of the Audio–Visual Fusion Pipelines

5 Evaluation Plan, Expected Outcomes and Deliverables

5.1 Evaluation Plan

The proposed system will be evaluated using the actor-independent training, validation, and test partitions. The same partitions will be maintained across the audio, visual, and multimodal experiments to ensure a consistent and fair comparison. This allows the system to be evaluated on actors who were not seen during training.

The audio, visual and multimodal emotion-recognition approaches will be evaluated using the same classification metrics, enabling comparison of different feature representations and deep-learning feature extraction approaches within each modality, as well as the multimodal fusion strategies.

The following classification metrics will be used for classification evaluation:

- **Accuracy:** Measures the overall proportion of correctly classified emotion samples.
- **Precision:** Measures how reliably samples predicted as a particular emotion actually belong to that emotion.
- **Recall:** Measures how well the system identifies samples belonging to each emotion class.
- **F1-score:** Provides a balanced measure of precision and recall.
- **Macro-averaged F1-score:** Calculates the F1-score for each of the eight emotion classes and averages them equally, ensuring that every emotion contributes equally to the evaluation.
- **Confusion Matrix:** Identifies which emotion classes are correctly recognized and which emotions are commonly confused with one another.
- **Repeated Actor-Independent Evaluation (if feasible):** Experiments may be repeated using different actor-independent splits. If performed, the mean and standard deviation of the evaluation metrics will be reported to assess the average performance and consistency of the system across different unseen actors.

The evaluation will compare different feature representation techniques and multimodal fusion strategies for emotion classification.

For the audio modality, handcrafted audio features and features learned using deep-learning-based feature extractors will be evaluated individually and in combination. Different suitable deep-learning architectures will be investigated for feature extraction, and the fused feature representations will be passed to an MLP classifier for eight-class emotion classification. The different deep-learning architectures will be compared based on the final audio classification performance to identify an effective audio feature extraction approach.

For the visual modality, handcrafted visual features and features learned using deep-learning-based feature extractors will also be evaluated individually and in combination. Different suitable deep-learning architectures will be investigated for visual feature extraction, and the fused feature representations will be passed to an MLP classifier for eight-class emotion classification. The different deep-learning architectures will be compared based on the final visual classification performance to identify an effective visual feature extraction approach.

The selected audio and visual representations will subsequently be evaluated using the multimodal fusion strategies proposed in Section 4, including decision-level fusion, feature concatenation, gated fusion and, if computationally feasible, cross-attention fusion. The

performance of these fusion strategies will be compared using the defined classification metrics.

The multimodal results will also be compared with the corresponding unimodal audio and visual results to assess whether combining both modalities improves emotion-recognition performance. Model configurations and fusion parameters will be selected using the validation set, while the held-out test set will be reserved for final performance evaluation.

5.2 Expected Outcomes

- **Deep Feature Extraction Comparison:** Determine how different deep-learning feature extraction approaches affect the final classification performance of the audio and visual modalities.
- **Feature Representation Assessment:** Determine whether fused handcrafted and deep-learned features improve emotion classification compared with handcrafted-only or deep-learned-only representations.
- **Multimodal Fusion Comparison:** Determine how the proposed audio-visual fusion strategies affect emotion-recognition performance.
- **Unimodal vs. Multimodal Evaluation:** Determine whether combining audio and visual modalities improves performance compared with the individual modalities.
- **Final System Selection:** Identify the final audio feature extractor, visual feature extractor, feature representation and multimodal fusion strategy based on the experimental results.

5.3 Expected Deliverables

The expected project deliverables include:

- Preprocessed and paired RAVDESS audio and video data with actor-independent training, validation and test partitions.
- Audio emotion-recognition pipeline incorporating handcrafted features, deep-learned features, feature fusion and MLP-based emotion classification.
- Visual emotion-recognition pipeline incorporating handcrafted features, deep-learned features, feature fusion and MLP-based emotion classification.
- Multimodal emotion-recognition pipeline supporting the fusion strategies proposed in Section 4.
- Experimental evaluation results covering the audio, visual and multimodal approaches using the defined classification metrics.
- External sample evaluation using manually labelled unseen audio-video samples to provide a preliminary assessment beyond the RAVDESS dataset.
- A lightweight demonstration interface for multimodal emotion prediction.

5.4 External Sample Evaluation and Demonstration Interface

The final multimodal system will be tested on a small set of manually labelled external audio-video samples not included in RAVDESS. The predicted emotions will be compared with the assigned labels to provide a preliminary assessment of the system's performance on unseen data and under different recording conditions.

A lightweight demonstration interface will allow users to provide an audio-video sample for emotion recognition. The selected audio and visual processing pipelines and multimodal fusion method will generate the final emotion prediction, which will be displayed together with the corresponding confidence scores where feasible.

6 Future Work

Future extensions of the proposed system may include:
- **Broader Emotion Recognition:** Extend the system beyond the eight RAVDESS speech emotions by incorporating suitable datasets containing additional emotion categories.
- **Robustness to Unreliable Modalities:** Investigate system performance under challenging conditions such as noisy audio, poor lighting, facial occlusion or reduced video quality.
- **Emotion Recognition in Songs:** Investigate the RAVDESS song subset separately to study emotion recognition in sung expressions.
- **Real-Time and Efficient Inference:** Explore frame-reduction strategies, batch processing, model compression, quantization and lightweight architectures to reduce computational requirements and support faster or real-time emotion recognition.
- **Unsupervised Emotion Pattern Discovery:** Explore approaches such as PCA and K-Means to identify hidden patterns and natural groupings within extracted emotion-related features. Cluster visualization and measures such as the Silhouette Score may be used for evaluation.
- **Extended Model and Hyperparameter Exploration:** Investigate additional machine-learning and deep-learning architectures, feature extractors and hyperparameter configurations for the audio and visual modalities to further improve emotion-classification performance.

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